

GBIF Backbone Taxonomy & REST API Specification

GBIF Secretariat (2026)

Global Biodiversity Information Facility Backbone Dataset

Checklist DOI: 10.15468/39omei | API Endpoint: <https://api.gbif.org/v1/>

Overview & Taxonomic Architecture

The GBIF Backbone Taxonomy is a single, synthetic management hierarchy used to organize species occurrence records across global biodiversity datasets. It integrates major biological classification databases including Catalogue of Life, WoRMS (World Register of Marine Species), and ITIS.

Key API Endpoints Used in AquaPulse AI Process

1. **Species Match API:** GET `/v1/species/match?name={species_name}` resolves informal or local species names to authoritative scientific taxonomy (Kingdom, Phylum, Class, Order, Family, Genus, Species) with confidence metrics.
2. **Occurrence Media API:** GET `/v1/occurrence/search?scientificName={name}&mediaType=StillImage` retrieves high-resolution field photos of marine species for visual reference.

Local Disk Memory Caching

AquaPulse caches fetched taxonomy and reference imagery into `species_snapshots/wiki_cache/` to eliminate latency, prevent API rate-limiting, and enable full offline operation during field telemetry deployments.